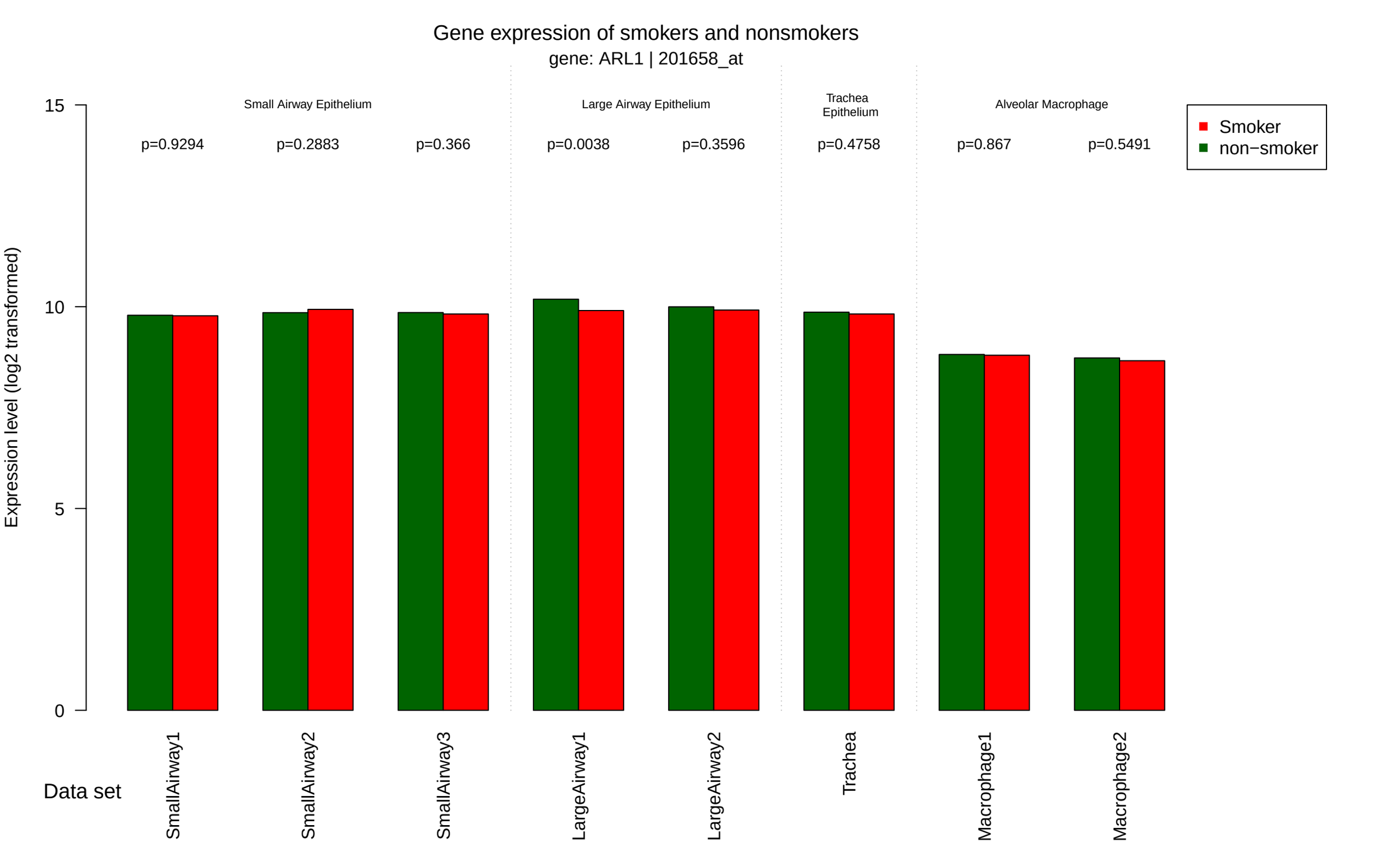
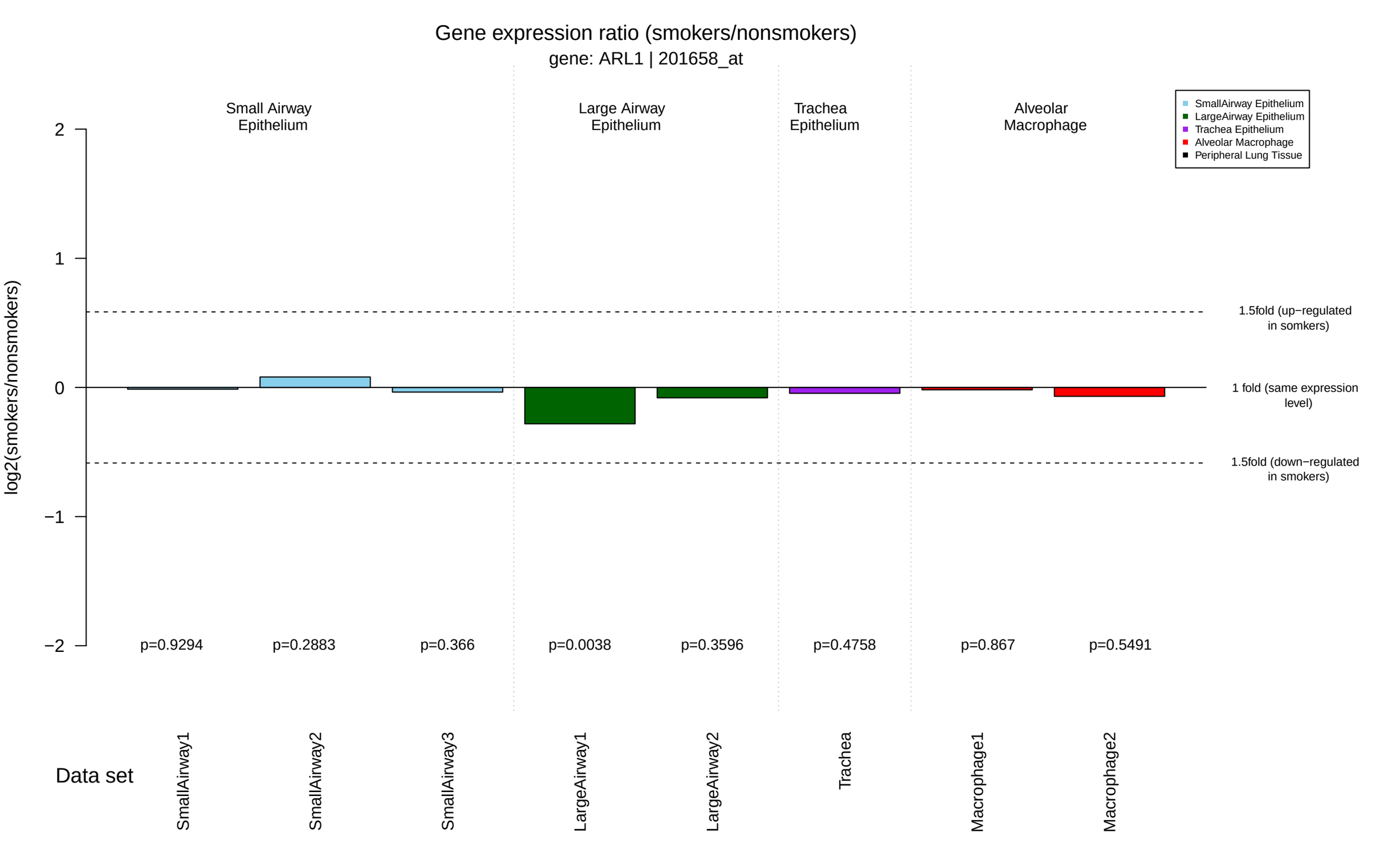
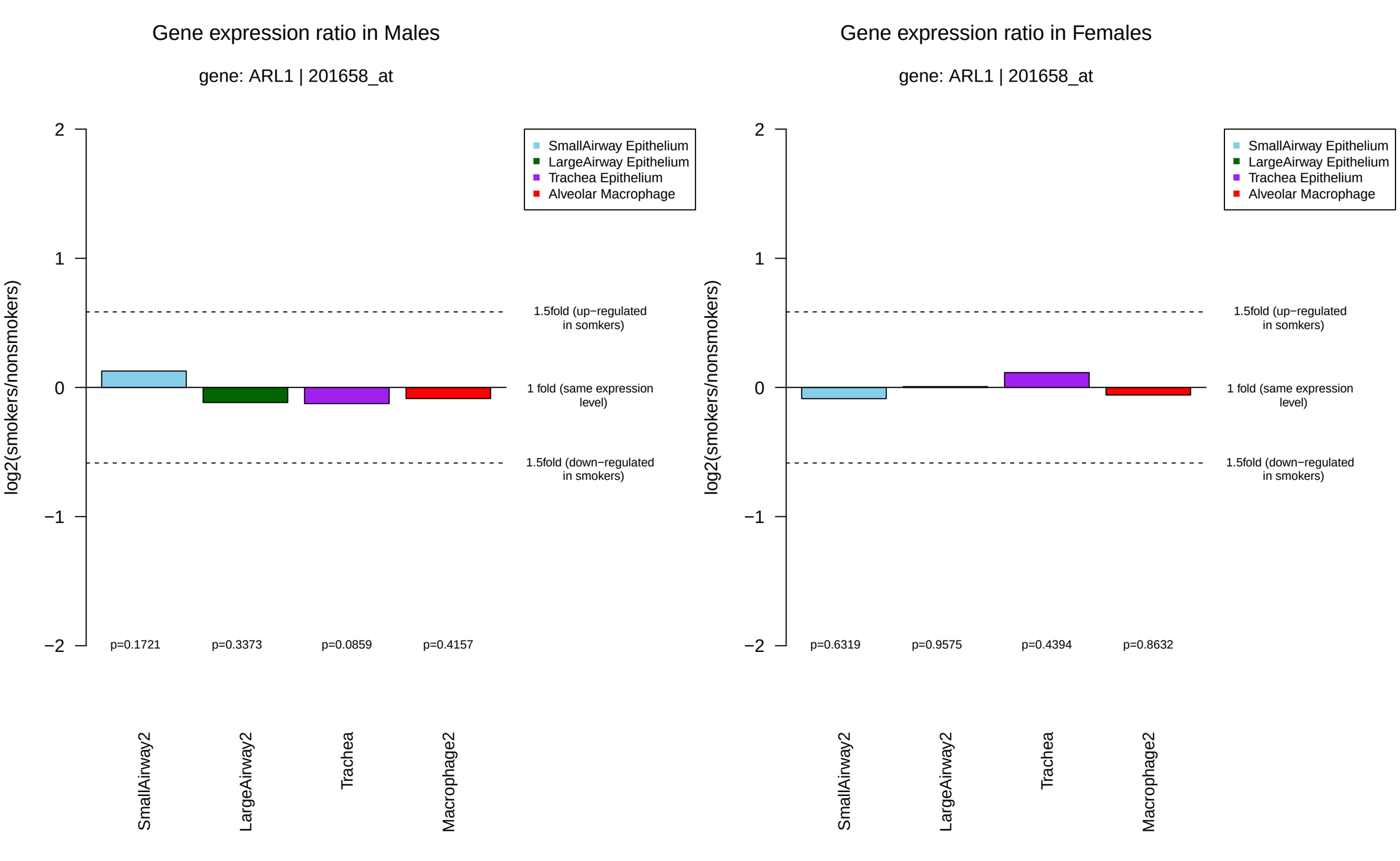
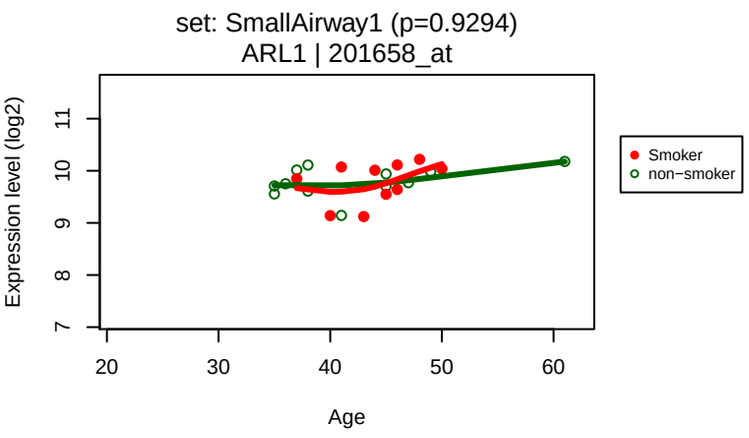
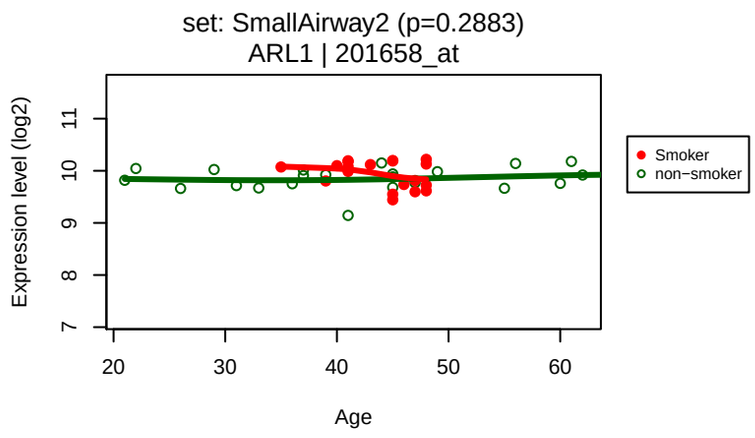
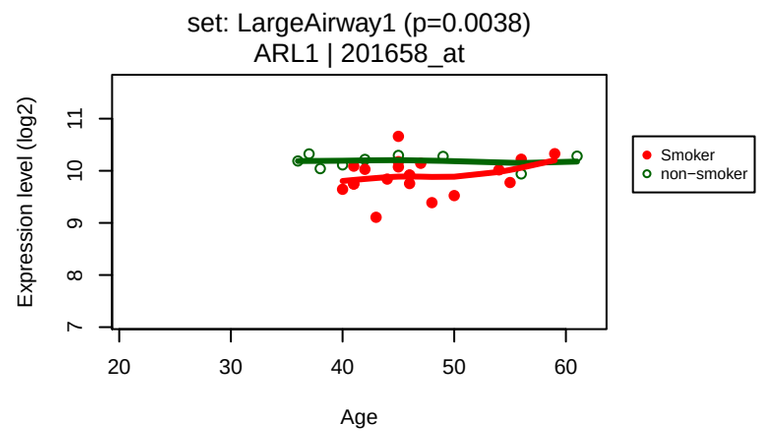

Gene expression of smokers and nonsmokers
gene: ARL1 | 201658_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9294
p=0.2883
p=0.366
p=0.0038
p=0.3596
p=0.4758
p=0.867
p=0.5491
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

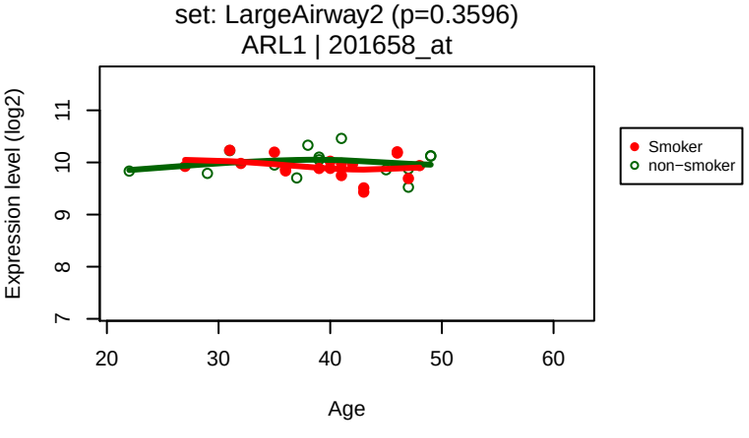

Gene expression ratio (smokers/nonsmokers)
gene: ARL1 | 201658_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9294
p=0.2883
p=0.366
p=0.0038
p=0.3596
p=0.4758
p=0.867
p=0.5491
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

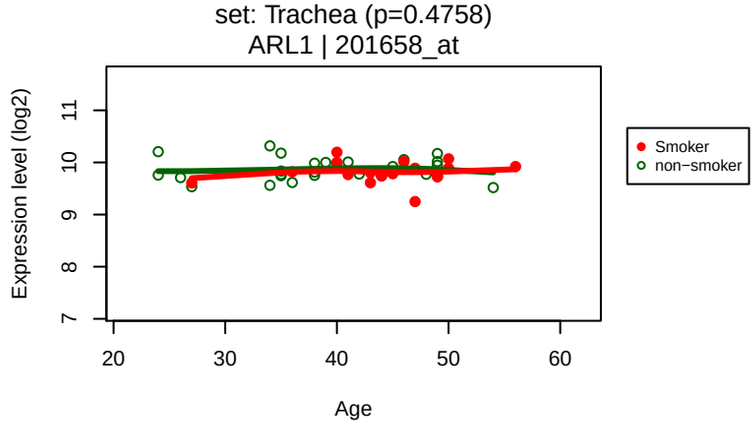

Gene expression ratio in Males
gene: ARL1 | 201658_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1721
p=0.3373
p=0.0859
p=0.4157
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL1 | 201658_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6319
p=0.9575
p=0.4394
p=0.8632
SmallAirway2
LargeAirway2
Trachea
Macrophage2

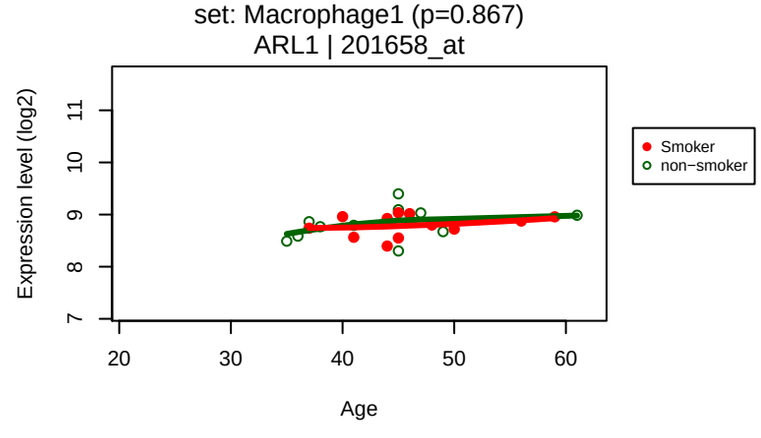

set: SmallAirway1 (p=0.9294)
ARL1 | 201658_at
Expression level (log2)
Age
Smoker
non−smoker

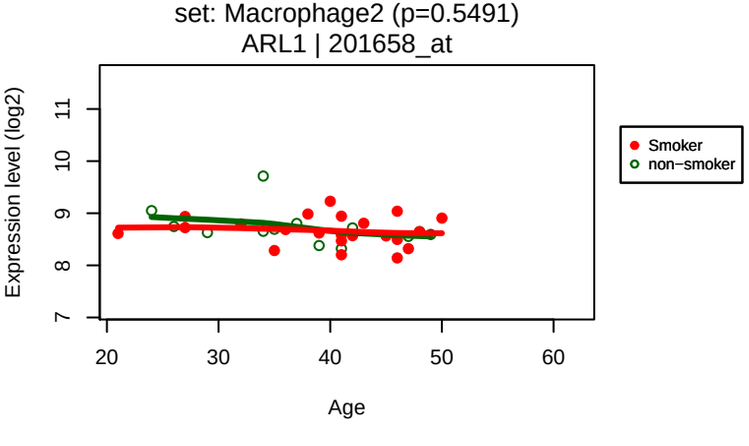

set: SmallAirway2 (p=0.2883)
ARL1 | 201658_at
Expression level (log2)
Age
Smoker
non−smoker

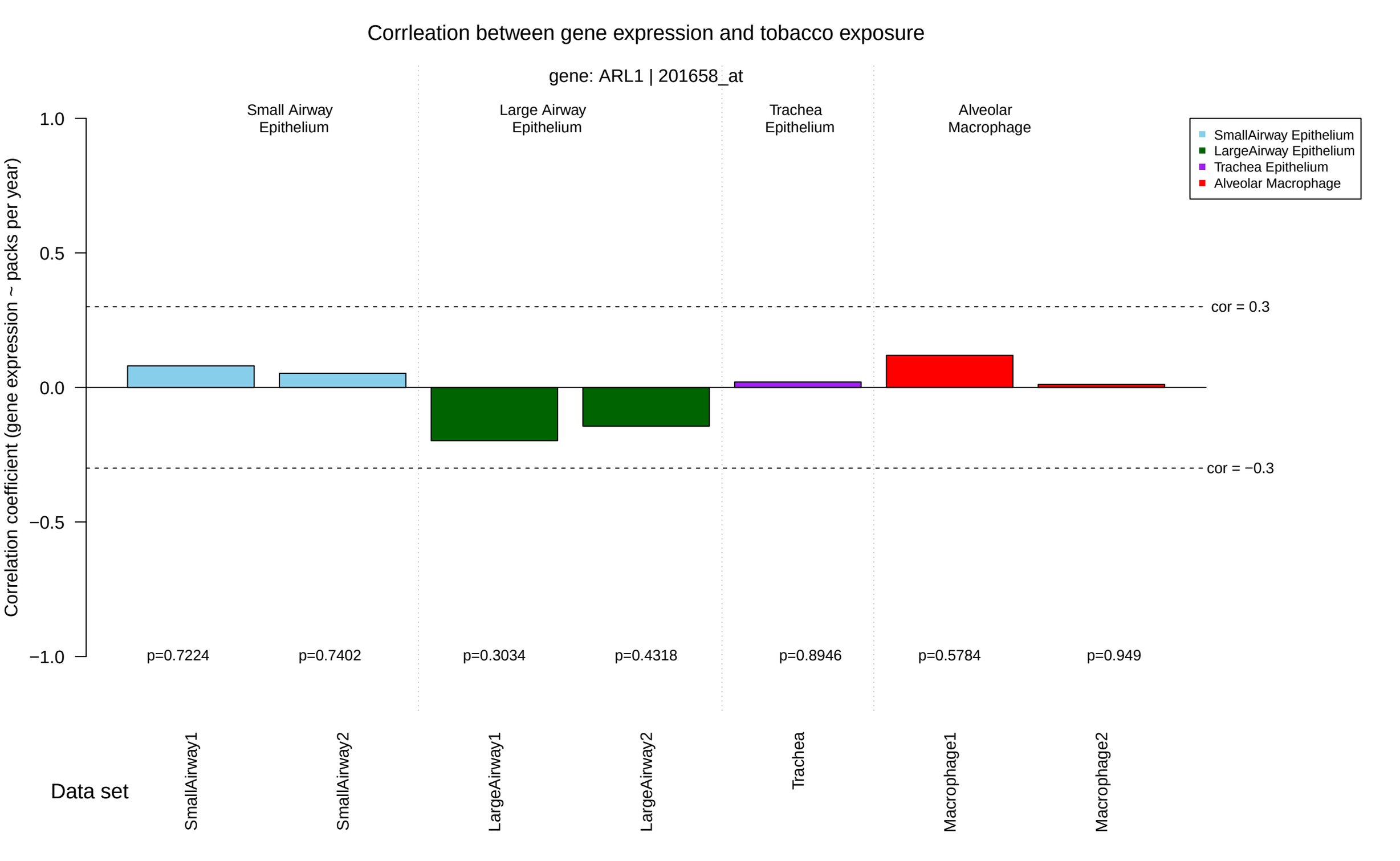

set: LargeAirway1 (p=0.0038)
ARL1 | 201658_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3596)
ARL1 | 201658_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4758)
ARL1 | 201658_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.867)
ARL1 | 201658_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5491)
ARL1 | 201658_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL1 | 201658_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7224
p=0.7402
p=0.3034
p=0.4318
p=0.8946
p=0.5784
p=0.949
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2